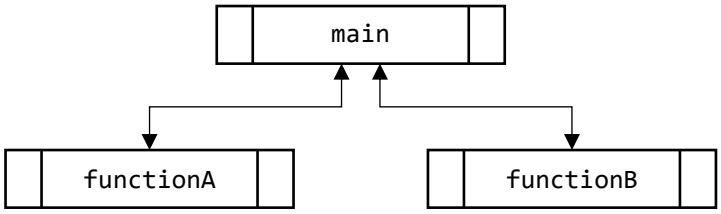
ECE 2305 – Introduction to C Programming
Programming Tutorial 18
User-defined Functions and Function Calls

Write a C++ program to demonstrate user-defined functions and function calls as illustrated below.



The functions are described below in "pseudo code."

functionA (local variable *n*)
    n = *n* + 1
    display the number of times functionA has been called
    display Memory Map for functionA
        variable *n* address and value
    return the value of *n*

functionB (local variable *n*)
    n = *n* + 1
    display the number of times functionB has been called
    display Memory Map for functionB
        variable *n* address and value
    return the value of *n*

main function:
    declare: variable *a* = the number of times functionA is called
    declare: variable *b* = the number of times functionB is called
    menu for main
        display Memory Map for function Main
            variable *a* address and value
            variable *b* address and value
        1. call functionA, send the value of *a*
            assign the return value to *a*
        2. call functionB, send the value of *b*
            assign the return value to *b*
        3. exit

Document the exercise in a Word document.

ECE 2305 – Introduction to C Programming
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Input

```
// ECE 2305 - Tutorial 18
#include <iostream>
using namespace std;

int functionA(int n) {
    static int countA = 0;
    countA++;
    n = n + 1;
    cout << "Function A called " << countA << " times." << endl;
    cout << "Memory Address of n in functionA: " << &n << " Value: " << n <<
endl;
    return n;
}

int functionB(int n) {
    static int countB = 0;
    countB++;
    n = n + 1;
    cout << "Function B called " << countB << " times." << endl;
    cout << "Memory Address of n in functionB: " << &n << " Value: " << n <<
endl;
    return n;
}

int main() {
    int a = 0, b = 0, choice;

    while (true) {
        cout << "\nMemory Map for Main Function:" << endl;
        cout << "Variable a Address: " << &a << " Value: " << a << endl;
        cout << "Variable b Address: " << &b << " Value: " << b << endl;

        cout << "\nMenu:" << endl;
        cout << "1. Call functionA" << endl;
        cout << "2. Call functionB" << endl;
        cout << "3. Exit" << endl;
        cout << "Enter your choice: ";
        cin >> choice;

        if (choice == 1) {
            a = functionA(a);
        }
        else if (choice == 2) {
            b = functionB(b);
        }
        else if (choice == 3) {
            cout << "Exiting program..." << endl;
            break;
        }
        else {
            cout << "Invalid choice, please try again." << endl;
        }
    }
    system("pause");
    return 0;
}
```

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Output

